

Muhammad Fakhrrur Rozi

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Central Java, Indonesia

Summary

Self-taught in computer science during and after an Arabic Literature degree, driven by a fascination with formal languages and type systems. Functional programmer with deep experience in Haskell and Koka, actively learning Scala 3. Passionate about type systems, compilers, language design, category theory, and purely functional design. Seeking a junior role where I can contribute to production FP systems and grow alongside a strong engineering team.

Education

Sarjana Sastra Arab (Arabic Literature)

2021 – 2025

Ma'had Aly Andalusia Banyumas

GPA: 3.65 / 4.0 — Self-taught in CS during and after this degree, driven by fascination with formal languages and type systems

SMA Islam Andalusia Kebasen

2017 – 2020

Banyumas, Indonesia

Skills

Primary Languages Haskell, Koka, OCaml, Rust, TypeScript/JavaScript

Familiar With Go, Idris2, Agda, Lean, Common Lisp, Nushell

Haskell Ecosystem servant, warp, aeson, megaparsec, lens, QuickCheck, hedgehog, streaming, polysemy, hspec, tasty, req, hasql, relude

Tools Git, Linux (NixOS, Arch), Neovim, Cabal, GHCup, HLS, Stack, Nix, hledger, Node.js, Playwright

Soft Skills Self-directed learning (15+ languages independently), technical communication, teaching & mentoring

Projects

derive — Koka Derive CLI

github.com/LitFill/derive

CLI tool eliminating boilerplate by auto-deriving `+show+`, `+eq+`, and typeclass-like functions in Koka, similar to GHC's `+deriving+`. Supports 2 derivation kinds, file I/O, and dry-run mode. Built because Koka lacked metaprogramming — solved a real workflow pain point.

sina — Hindley-Milner Type Inference

github.com/LitFill/sina

Full HM type inference (Algorithm W) implemented from scratch in Haskell: parser, AST, constraint-based type checker with let-polymorphism. Supports integers, booleans, conditionals, let-bindings, and binary ops.

klens — Value-based Lenses in Koka

github.com/LitFill/klens

Value-based lenses for Koka: one-shot focused views with `+set+`, effectful `+modify+` propagating Koka effects, composition, and tuple optics — a design exercising Koka's effect system.

src-todo — TODO Tracker CLI

github.com/LitFill/src-todo

Haskell CLI managing TODO comments in source files: register, list, and unregister todos with unique UUIDs, updated in place. Comment parsing built with Megaparsec.

Experience

Secretary / Administrative Automation Yayasan Al Anwar Al Hisyamiyyah — Pondok Pesantren 2020 – Present

- Built Playwright automation scripts to batch-enter student data for 3,000+ santri into the EMIS (Education Management Information System), eliminating manual data entry entirely
- Developed Node.js docx generators and OOXML manipulation tools to produce Laporan Pertanggungjawaban (accountability reports) for Pondok management, reducing preparation time from days to minutes

- Designed document automation workflows combining Playwright, OOXML, and custom scripts to streamline administrative reporting across 10+ report types

Programming Instructor Extracurricular — Ma'had Aly Andalusia Banyumas

First half 2026

- Taught programming fundamentals to students with no prior coding experience — 10+ students over 6 months
- Designed curriculum and hands-on exercises from scratch, adapting content for absolute beginners with no CS background
- Developed teaching materials and guided students through practical coding sessions

Open Source Contributions

- BNFC (BNF Converter) — Reported and fixed extraneous closing bracket in string literal doc generation
- Noctalia Shell — Custom plugins for the Noctalia desktop environment

Certifications

English Language Certification — BLK, Kementrian Ketenagakerjaan (Ministry of Manpower)

Languages

English (professional working), Indonesian (native), Arabic (academic)